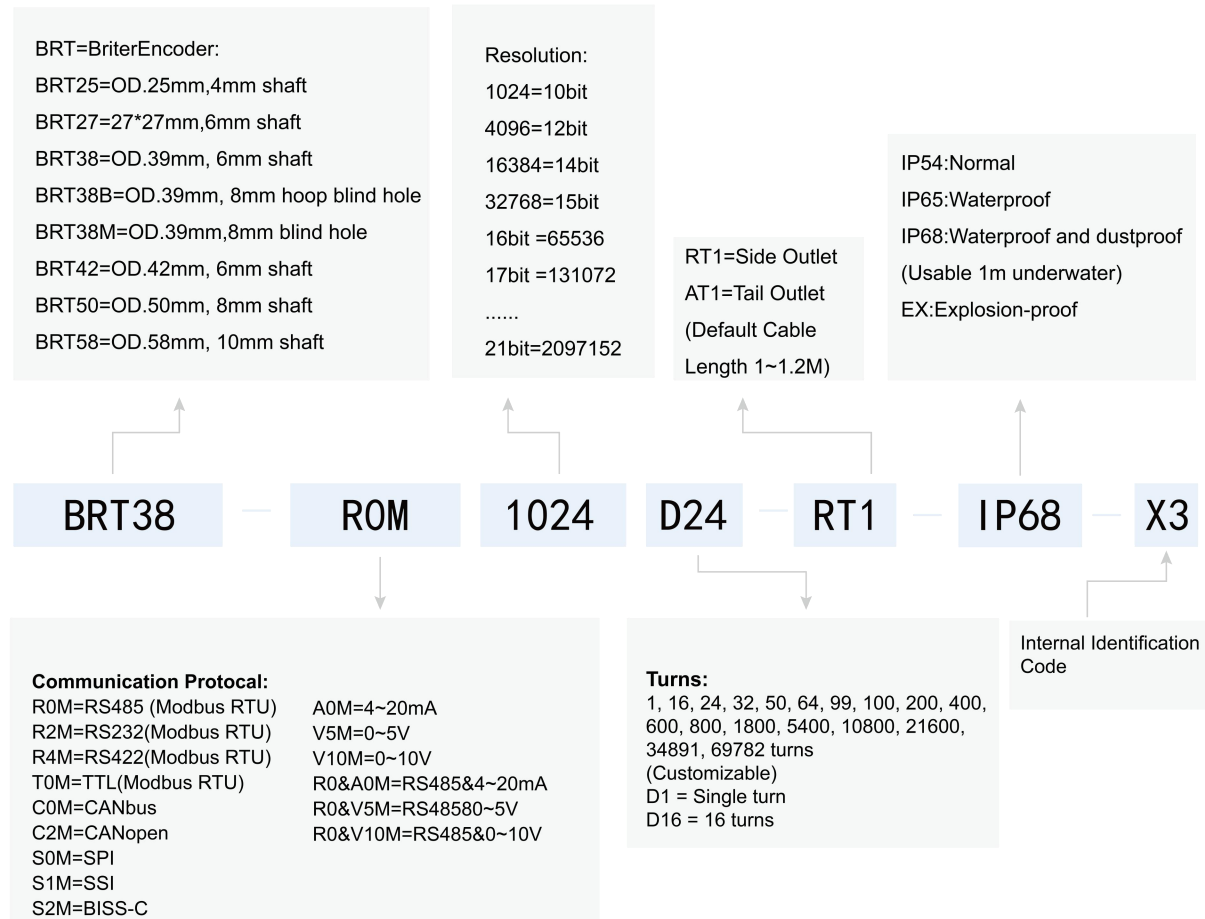


# Briterencoder Absolute Rotary Encoder

## Analog 4~20mA Current Output User Manual

### 1. Product Model Description



Note 1: For photoelectric types, add P before the communication interface, e.g., photoelectric RS485 = PROM.

Note 2: For high-speed RS485 communication interfaces compatible with Tamagawa models, add HS before the communication interface, e.g., high-speed RS485 = HSR0M.

Note 3: For analog speed models, note the maximum speed RPM after the model, e.g., BRT38 - A0M1024 - RT1 - IP68(1000RPM).

#### Model Description:

- 1.Model:e.g., BRT25 indicates an outer diameter of 25mm and an output shaft of 4mm.
- 2.Communication interface: e.g., ROM means the electrical interface is RS485 and the communication protocol is Modbus RTU.
- 3.Resolution: It refers to the single - turn resolution, which has nothing to do with the subsequent number of turns. e.g., with a 10 - bit resolution, 2 to the power of 10 equals 1024. That is, the 360° full circle is divided into 1024 parts, and the minimum angular resolution is  $360^\circ/1024 = 0.38^\circ$ .
- 4.Turns:It represents the power-off memory range, rather than the mechanical rotation turns of the encoder. The power-off memory of a single-turn encoder is limited to one turn, while a multi-turn encoder can record and restore multi-turn position information. Both types of encoders can perform infinite mechanical rotations.
- 5.Blind holes are mainly used to provide space for fixed threaded connections, while hoop blind hole are mainly used to fix and connect components. In practical applications, it is more recommended to use hoop blind hole for encoders.
- 6.Inventory: Some randomly combined models may be out of stock. Please consult in advance to ensure that the selected model is available.

## 2. Wiring Definition and connection

|                       |                  |                                                                                                                                                                                                                                                                           |
|-----------------------|------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Grey wire             | 4-20mA positive  | -                                                                                                                                                                                                                                                                         |
| Brown wire (optional) | 4-20mA negative. | 1. In three-wire configuration, this line remains unconnected as the signal negative and power negative are directly connected.<br>2. In four-wire configuration, this line needs to be connected to the negative of the 4-20mA current loop (i.e., the signal negative). |
| Black wire            | 0V (GND)         | -                                                                                                                                                                                                                                                                         |
| Red wire              | Power Supply     | DC 12-24V                                                                                                                                                                                                                                                                 |
| Orange wire           | Function line    | SETH/ Direction                                                                                                                                                                                                                                                           |
| Yellow wire           | Function line    | SETL/ Reset                                                                                                                                                                                                                                                               |

## 3. Product Parameters

|                           |                                                                                       |                          |                                                       |
|---------------------------|---------------------------------------------------------------------------------------|--------------------------|-------------------------------------------------------|
| Multi-turns               | Singleturn, 16, 25, 32, 50, 64, 100, 400, 1800, 5400, 10800, 21600turns(Customizable) |                          |                                                       |
| Resolution                | 10bit, 12bit, 13bit, 14bit, 15bit, 16bit                                              |                          |                                                       |
| Supply Voltage            | 12~24V                                                                                | Linearity                | 0.1%                                                  |
| Kernel Refresh Period     | 50uS                                                                                  | Electrical lifespan      | > 100000 h                                            |
| WorkingCurrent            | 100mA                                                                                 | Protection Level         | IP54, IP68, EX                                        |
| Shell/ Flange Material    | Galvanized steel / Aluminum alloy<br>IP68/EX Stainless Steel                          | Maximum mechanical speed | Singleturn 8000RPM<br>Multi-turn 6000PRM              |
| Shaft Material            | Stainless Steel(Customizable)                                                         | Starting Torque          | 0.006Nm                                               |
| Bearing material          | Bearing Steel                                                                         | Weight                   | 86~200 g ((Includes 1.2m shielded wire, Customizable) |
| Maximum load on the shaft | Axial 20 N, Radial 80 N                                                               | Humidity                 | 98 % ( No condensation)                               |
| Working temperature       | -40 ~ + 85 °C                                                                         | Storage temperature      | -40 ~ +85°C                                           |

### How to calculate the encoder rotation angle through sensor feedback analog signal?

Determine the number of turns and signal output type of the sensor you purchased.

#### Output 4-20mA encoder:

The angle=number of turns \* 360 \* (current -4)/16 (unit: °)

E.g. If a single turn 4-20mA encoder is used and the current is 12mA, then the angle=1 \* 360 (12-4)/16=180 °

## 4. Function Line Setting Method

#### Set the minimum value:

When power on, use the yellow wire to touch the black wire (0V) for a short time (more than100ms) to set the current position to the minimum analog value;

#### Set the maximum value:

When power on, use the orange wire to touch the black wire (0V) for a short time (more than100ms) to set the current position to the maximum analog value;

### Set the midpoint value:

When power on, use the yellow wire and the orange wire for to touch the black wire (0V) a short time (more than 100ms) to set the current position to the midpoint analog value;

### Restore the encoder to factory settings:

When the encoder is in power-off state, connect the yellow wire to the black wire (0V). Then power on and keep it on for 2 minutes (110~130 seconds) to reset the encoder.

### Set the encoder's value direction

When the encoder is in a powered-off state, the orange wire contacts the black wire (0V). Then power on and keep it on for 2 minutes (110~130 seconds), the encoder can switch its direction of value.

## 5. Precautions and warranty

- Encoders belong to precision instruments. Please handle them with care and use them with care, especially do not knock, hit, or forcefully pull the encoder shaft.
- The encoder and mechanical connection should use flexible connectors or elastic brackets to avoid hard damage caused by non concentric rigid connections.
- Although the encoder itself does not lose the number of turns in interference environments, it can cause interference to the data during transmission. Therefore, when there is a motor or strong electromagnetic interference environment in the system, an isolated power supply should be used to power the encoder. And when there's external extended communication lines, it is best to use double shielded cables.
- The encoder casing and shielding wire should be well grounded to prevent damage to the encoder circuit caused by lightning strikes or high-voltage static electricity
- The product is guaranteed for one year free of charge when used correctly.
- When exceed the warranty period, or the product is damaged due to improper use, the product can be sent back to the original factory for repair (only raw material cost is required when repair).

## 6. Encoder Indicator Lights Description

1. The status light displays are as follows:

1) Default indication status: "Blue -> Blue -> Blue -> Cyan -> Blue" 1s slow flash, indicating normal power supply to the encoder;

2) Other work status indications:

- a. Setting minimum value status: Orange light flashes quickly at 0.5s intervals;
- b. Setting maximum value status: Purple light flashes quickly at 0.5s intervals;
- c. Setting midpoint value status: Orange light flashes quickly at 0.5s intervals;
- d. Power-on reset status: Orange light flashes quickly at 0.5s intervals;
- e. Setting direction status: Purple light flashes quickly at 0.5s intervals;

3) Red light indicates encoder failure.

## 7. Contact us and technology support

WhatsApp/Wechat: +86 15183276844

Email: [brt@briterencoder.com](mailto:brt@briterencoder.com)

### Technical documentation

Detailed version of the instruction manual;  
PC software;  
2D drawings and 3D model files;  
Additional Video Tutorials;

For more details, please visit our website:  
[www.briterencoder.com](http://www.briterencoder.com).



WhatsApp



WeChat